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Factors Contributing to Interindividual Variation in Retronasal Odor Perception from Aroma Glycosides: The Role of Odorant Sensory Detection Threshold, Oral Microbiota, and Hydrolysis in Saliva

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Supporting Information

ABSTRACT: Glycosides are sugar conjugates of aroma compounds that are found in many fruits and vegetables, and while glycosides are non-volatile, they can release flavor during eating, through enzyme hydrolysis from oral microbiota. Recently, a range of sensory phenotypes for glucoside perception have been observed, reflecting interindividual variation in response to precursors of floral and smoky flavors, geranyl glucoside and guaiacyl glucoside. To understand this variation and investigate the role of oral microbiota on *in vitro* hydrolysis of glucosides in saliva, metagenomic screening was conducted using individuals representing the range of sensory phenotypes for geranyl and guaiacyl glucosides. In parallel, sensory retronasal detection thresholds for geranyl glucoside, guaiacyl glucoside, and the volatile odorants geraniol and guaiacol were determined. Oral microbial communities correlated with hydrolysis of glucosides in saliva, but the relationship did not extend to sensory phenotypes. Overall, the retronasal detection threshold of the volatile odorants studied was the main factor determining sensory phenotype.

KEYWORDS: *glycosides, sensory detection threshold, oral microbiota, hydrolysis, saliva, intraindividual variation*

INTRODUCTION

Glycosides are non-volatile odor precursors that have no odor themselves but can release odorants upon hydrolysis. In wine, glycosides are an important source of varietal wine flavor; they originate in the grape berry, are extracted during crushing and fermentation, persist in wine, and can release varietal odor upon enzyme or acid hydrolysis.[1−3] The sugar component of glycosides can consist of many different types of monosaccharides and disaccharides, and research has often focused on glucosides, which consist of glucose as the sugar moiety. While hydrolysis occurs relatively slowly during wine processing and storage, hydrolysis of glycosides can occur very rapidly in the human mouth, similar to the release of potent thiols from cysteine conjugates.[4] The mechanism of flavor perception from intact glycosides requires the release of the volatile odorant by glycosidase enzymes within seconds of placing the glycoside in the mouth and then is followed by retronasal olfaction of the volatile odorant.[5,6] This phenomenon has been demonstrated for hexyl glucoside, geranyl glucoside, and glycosides of volatile phenols and other monoterpenes, although not all individuals tested were able to perceive flavor from the glycosides.[5−7]

Recently, significant interindividual variation in glycoside flavor perception was highlighted in a study of three different glycosides: guaiacyl glucoside, geranyl glucoside, and glycosides extracted from Gewürztraminer.[6] A number of sensory phenotypes were observed, and for each glucoside, individuals were classified as "tasters" or "non-tasters" depending upon whether they perceived flavor from that glucoside. Some individuals perceived flavor from all three of the glycosides tested (28% of individuals); some did not perceive flavor from any of the three glycosides (23% of individuals); and others responded to only one or two of the glycosides tested.[6] Considering only pure geranyl glucoside and guaiacyl glucoside, four phenotypes were observed: "taster" of both glucosides, "non-taster" of both glucosides, "taster" of geranyl glucoside but "non-taster" of guaiacyl glucoside, and "non-taster" of geranyl glucoside and "taster" of guaiacyl glucoside.

There are many factors that potentially can contribute to interindividual variation in the sensory response to glycosides. Oral microbiota are fundamentally important because they produce the majority of glycosidase enzymes in the mouth, and oral microbiota vary between individuals.[8−10] Another important factor is the extent of enzyme hydrolysis of the glycoside, which also varies widely between individuals.[11] The third critical factor is the ability of individuals to detect and recognize the odorants. Olfactory detection thresholds are a

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known source of variation. Although they are usually reported as a single value (group best estimate threshold), they commonly vary between individuals by 3 orders of magnitude or more and typically 14−23% of individuals are insensitive to a given odor.[12] Other plausible factors that might contribute to interindividual variation in flavor perception from glycosides include variations in the saliva flow rate and buffering capacity and mouth movement and breathing behavior.[13]

The presence of microbial glycosidase enzymes in saliva was demonstrated as early as 1954,[14] with ramifications for bioactivity of natural products in the diet as well as medication and dental health. As a result of the far reaching implications, extensive research has been performed in this area, ranging from general studies on the bacterial origin of salivary enzymes to studies showing specific β-glucosidase enzyme activity of oral bacterial isolates, including species belonging to the genera *Streptococcus* and *Prevotella*.[15,16]

Recent advances in DNA sequencing technologies have helped to elucidate the extent of the oral microbial community diversity, unravelling a more complex ecosystem of bacteria than previously thought.[17] Despite the complexity of this system, the microbial community appears to be stable over time and similar community types have been observed in several metagenomic studies.[18−22] For example, metabolome assessment of the saliva of 268 healthy young adults revealed a microbiota−metabolite association, suggesting two different ecological states: one functionally adapted to proteolysis and amino acid fermentation and another group adapted for saccharolysis.[18] It is unclear whether these metabolic groups and specific bacterial species are associated with the production of extracellular β-glucosidases, because this specific trait is commonly observed under culture conditions in several bacterial species from the oral microbiome[15,16,23] and is predicted from the genome for a range of oral bacterial isolates.[24] Furthermore, fluctuations in parameters, such as pH, are related to these oral ecological states, possibly affecting the *in situ* activity of the bacterial β-glucosidases.

Limited research has focused on investigating the oral microbial communities associated with β-glucosidase activity and the release of volatiles from flavor precursors. However, techniques, such as 16S rRNA gene sequencing, can provide insights into the microbial communities associated with the release of these volatile compounds in the mouth environment, yet recent studies have only tested oral bacterial strains under culture conditions,[2,3,4] which do not reflect the time frame relevant to tasting or the conditions of the mouth environment, where microorganisms exist in complex relationships with the host and other microorganisms.[25]

A large degree of interindividual variation (20−80-fold) in saliva glycosidase activity has been observed using substrates, such as *p*-nitrophenol glucoside, non-volatile phenolic glucosides, and smoky volatile phenol glycosides.[8,10,11,26] The chemical structure has been shown to affect hydrolysis for both the glycoside and the aglycone moiety; this highlights the need to test salivary glycosidase activity with the specific glycosides used in sensory experiments as opposed to relying on the convenient and rapid *p*-nitrophenol glucoside as a model representing glycosidase activity in general. While it is accepted that glycoside hydrolysis in whole saliva must influence the perception of flavor from glycosides, the link between the extent of hydrolysis and sensory perception of glycosides has not been addressed to date.

On the basis of previous research into the interindividual sensory variation observed from pure key grape and wine glycosides, this study seeks to answer the key question: "what determines the ability of an individual to perceive flavor from glycosides?" Specifically, how do oral microbiota, glycoside hydrolysis in saliva, and sensory detection threshold of the released odorant determine the sensory response to two pure glucosides found in wine, which are precursors for floral and smoky flavors: geranyl glucoside and guaiacyl glucoside. The association between the sensory phenotype and oral microbiota was characterized with 41 individuals using a sensory study in parallel with metagenomic analysis of oral DNA. A subset of 23 individuals representing a range of sensory phenotypes was studied in more detail to assess the oral bacterial communities and *in vitro* hydrolysis of the two glucosides by whole fresh saliva, comparing tasters and non-tasters of the glucosides. On the same day, sensory retronasal detection thresholds for volatiles geraniol and guaiacol and non-volatiles geranyl glucoside and guaiacyl glucoside were determined. This approach allowed us to probe whether the inability to perceive flavor from a particular glucoside was due to inefficient hydrolysis of the glucoside or because of a high sensory retronasal detection threshold of the released free volatile. The relationships between oral microbiota, saliva glycosidase activity *in vitro*, sensory detection thresholds of geraniol, guaiacol, geranyl glucoside, and guaiacyl glucoside, and glucoside taster phenotype were explored to understand the main factors driving perception of flavor from glucosides.

MATERIALS AND METHODS

General. The Human Research Ethics Committee of the University of South Australia (Adelaide, South Australia, Australia) approved the methods of testing and data collection used in this study, approval number 0000035390.

Pure water from a Milli-Q system (Millipore, North Ryde, New South Wales, Australia) was used for all chemical and sensory experiments. Geranyl glucoside, guaiacyl glucoside, d_7-geraniol, and d_3-guaiacol were synthesized in-house.[6,27−29] Geraniol and guaiacol were purchased from Sigma-Aldrich Pty Ltd. (Castle Hill, New South Wales, Australia). Glacial acetic acid and Lichrosolv-grade solvents ethanol, methanol, and acetonitrile were purchased from Rowe Scientific (Lonsdale, South Australia, Australia). Acros Organics calcium chloride (96%), Invitrogen PureLink Microbiome DNA Purification Kits, and Qubit dsDNA BR Assay Kits were purchased from Thermo Fisher Scientific Pty Ltd. (Scoresby, Victoria, Australia).

Defining Sensory Phenotype. A total of 41 individuals (mean age of 42, range of 24−62 years, and 25 males) were screened for their flavor response to geranyl glucoside tested at 3.00 mg/L and guaiacyl glucoside tested at 0.500 mg/L dissolved in water, with a water blank, as described previously.[6] Each individual completed the study in a single day. Individuals were instructed to refrain from brushing their teeth or using mouthwash in the morning before the sensory test and to refrain from eating and drinking (apart from water) in the hour before the test. Prior to the sensory session, individuals smelled familiarization samples of 300 μg/L geraniol and 300 μg/L guaiacol in water. After a brief discussion about descriptors, all individuals agreed to use the terms "fruity/floral" and "smoky/medicinal" to describe the samples in the study. The intensity of the sensory attributes "fruity/floral", "smoky/medicinal", and "other" was rated on a 15 cm unstructured line scale marked with anchor points of "low" and "high" placed at 10 and 90% of the scale, respectively. Individuals were instructed not to smell the sample but take the whole 10 mL aliquot of sample into their mouth, swirl it around for 5 s, and then expectorate the sample. After a forced wait of 1 min to allow flavors to develop, the individuals were asked to rate intensity of "fruity/floral", "smoky/medicinal", or "other". Individuals were then offered a piece

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of plain rice cracker (Ceres Organics brand original unflavored rice cracker from a local supermarket) and water to rinse their mouth and forced to wait at least 2 min before moving to the next sample. A total of nine samples were assessed in each session, comprising of triplicates of the three samples in a complete block design. The sensory test was repeated over several days, with a total of 41 individuals.

Responses for each individual were subjected to analysis of variance (ANOVA, Fizz sensory acquisition software, version 2.51, Biosystems, Couternon, France), and for each of the two glucosides, individuals were classed as "taster" phenotype if the mean rating for the sample was significantly higher than the control water ($p < 0.15$). There were four phenotypes: those with a significant response to both glucosides, those with a significant response to geranyl glucoside but not guaiacyl glucoside, those with a significant response to guaiacyl glucoside but not geranyl glucoside, and those with no significant response to either glucoside. From these results, a subset of individuals representing the four phenotypes was selected for threshold testing. The individuals are numbered 1−23 throughout the manuscript.

Oral Microbiota 16S rRNA Gene Sequencing and Data Processing. DNA from saliva samples was extracted using a PureLink Microbiome DNA Purification Kit following the instructions of the manufacturer. Amplicon sequencing was performed using primers 341F and 805R, targeting the V3-4 region of the 16S rRNA gene and the Miseq (Illumina) platform (2 × 300 bp paired-end reads). Forward and reverse reads were trimmed for quality using Trimmomatic, version 0.38.[30] The trimmed forward and reverse reads were merged using FLASH, version 1.2.11,[31] then dereplicated, and formatted for use in the UPARSE workflow.[32] Taxonomy was assigned with QIIME[33] using the Human Oral Microbiome Database (HOMD), version 15.1.[24] The data are available at the National Center for Biotechnology Information (NCBI) under BioProject accession number PRJNA558132.

Community types based on the microbial community structure were estimated using Dirichlet multinomial mixture (DMM) modeling[34] on an operational taxonomic unit (OTU) abundance level for 41 samples. The number of groups was obtained using the lowest Laplace estimation score.

Prior to ordination and statistical analyses, OTUs with low abundance (<10 counts) and bacterial classification were filtered and then subsampled at equal depth. Ordination analyses and abundance boxplots were performed in R[35] using the ampvis, version 2.4.5,[36] vegan, version 2.5.4,[37] and phyloseq, version 1.26.1,[38] packages.

Correlation analyses on a subset of 23 samples was performed in R with the log-transformed metadata using the stats package, version 3.5.2,[35] and plotted using ggplot2, version 3.1.0.[39] Metadata values were fitted into the ordination plot using the envfit function in vegan,[37] and significance of fit was estimated over 999 permutations. Temporal variation between samples was analyzed using hierarchical clustering, with the hclust function in the stats package. The LDA effective size (LEfSe) biomarker discovery tool[40] was used with the OTU abundance table in a single class mode and the logarithmic LDA score threshold of 2 ($p < 0.05$).

***In Vitro* Saliva Hydrolysis Assay.** Whole fresh saliva was incubated with glucosides as reported previously.[26] Briefly, 10 mL of saliva was collected from sensory panellists immediately prior to undertaking sensory experiments, collected directly into sterile 50 mL plastic centrifuge tubes. Parafilm (Sigma-Aldrich Pty Ltd., Castle Hill, New South Wales, Australia) was provided to assist with salivation, and the whole fresh saliva was used within 1 h of sampling. For each individual, two replicate incubations of saliva with glucosides and a blank consisting of saliva with no added glucosides were performed. Saliva (3 mL) was incubated with 5 μg each of guaiacyl glucoside and geranyl glucoside for 30 min at 37 °C. The reaction was stopped by adding 2 mL of saturated $CaCl_2$ and mixed, and then 100 μL was removed for liquid chromatography−mass spectrometry (LC−MS) analysis of remaining glucosides based on the previously described method.[6] To the remaining saliva, 100 μL of internal standard containing 20 mg/L deuterated geraniol and guaiacol was added and the samples were then analyzed by solid-phase microextraction−gas chromatography−mass spectrometry (SPME−GC−MS) straight away, as described previously.[6] See the Supporting Information for method details.

Threshold Testing. Retronasal detection thresholds were determined for four compounds: geraniol, guaiacol, geranyl glucoside, and guaiacyl glucoside. Geranyl glucoside and guaiacyl glucoside were tested in pure water. Geraniol and guaiacol were tested in pure water with 0.4% alcohol (v/v), because the stock solutions had been prepared in pure ethanol and were added to water to achieve the required concentrations. Ethanol was added to a concentration of 0.4% alcohol (v/v) in the water blanks for the geraniol and guaiacol thresholds. The purity and concentration of the geraniol and guaiacol solutions were confirmed by GC−MS analysis.

Given the limited amount of material available, the test was conducted using a panel of 23 people representing the different phenotypes rather than using a smaller panel with more replications per person. To eliminate possible variation from 1 day to the next, individuals were required to complete threshold tests for the four compounds on 1 day. The threshold tests were held on 2 days, with half the panel completing the tests on each day. Saliva was sampled for hydrolysis experiments and DNA extraction on the same day, prior to sensory testing.

A three-alternative forced choice (3-AFC) test was carried out based on the procedure described in ASTM method E 679-04 for each of the four compounds. The individuals ($n = 23$, 14 males, age of 24−58, and mean age of 43 years) all had experience in tasting wine and sensory methods. Individuals were instructed to refrain from brushing their teeth or using mouthwash in the morning before the sensory test and to refrain from eating and drinking (apart from water) in the hour prior to the sensory test. Samples were presented in sets of three ISO standard glasses, with each set consisting of two water blanks and one sample with the added compound. Each individual was presented with a minimum of six concentrations in increasing 3-fold steps, ranging from 0.003 to 72.9 μg/L for geraniol and guaiacol, from 0.1 to 2430 μg/L for guaiacyl glucoside, and from 0.37 to 7290 μg/L for geranyl glucoside. Individuals were instructed to assess the samples one at a time, not smell the sample, but take the whole 10 mL aliquot into their mouth, swirl it around in the mouth for 5 s, and then expectorate. There was a forced wait of 1 min to evaluate the sample and write notes if desired, and then individuals were instructed to take a sip of water to rinse their mouth and forced to wait at least 1 min before moving to the next sample. On the basis of previous sensory experiments, the wait and rest periods were appropriate and necessary to experience released odors from the glucosides and to avoid carryover effects. After tasting three samples, individuals indicated which sample was different to the other two. Plain unflavored rice crackers and water were offered during 3 min forced rests between each 3-AFC set of three samples. A break of at least 10 min was enforced between compounds. Prior to the geraniol and geranyl glucoside threshold tests, the panellists smelled a familiarization sample of 300 μg/L geraniol. Following the threshold test for geraniol and then geranyl glucoside, there was a longer break of 45 min when the panellists were offered morning tea, coffee, and half an hour rest. Then, the panellists smelled a familiarization sample of 300 μg/L guaiacol, followed by the threshold test for guaiacol, 10 min rest, and then the threshold test for guaiacyl glucoside.

A total of 14 individuals had incorrect responses at the highest original concentrations tested for one or more compound, but further testing of higher concentrations was not undertaken as a result of the limited quantity of geranyl glucosides and guaiacyl glucoside available and the time required to repeat the test. The best estimate threshold (BET) of each individual was calculated by taking the geometric mean of the highest incorrect response and the lowest correct response after which all consecutive responses were correct. The panel best estimate was calculated by taking the geometric mean of the individual BETs.

One-way ANOVA was used to compare the detection thresholds for geranyl glucoside and geraniol between geranyl glucoside tasters ($n = 11$) and non-tasters ($n = 12$) using JMP, version 14.3.0, software. One-way ANOVA was also used to compare the thresholds for

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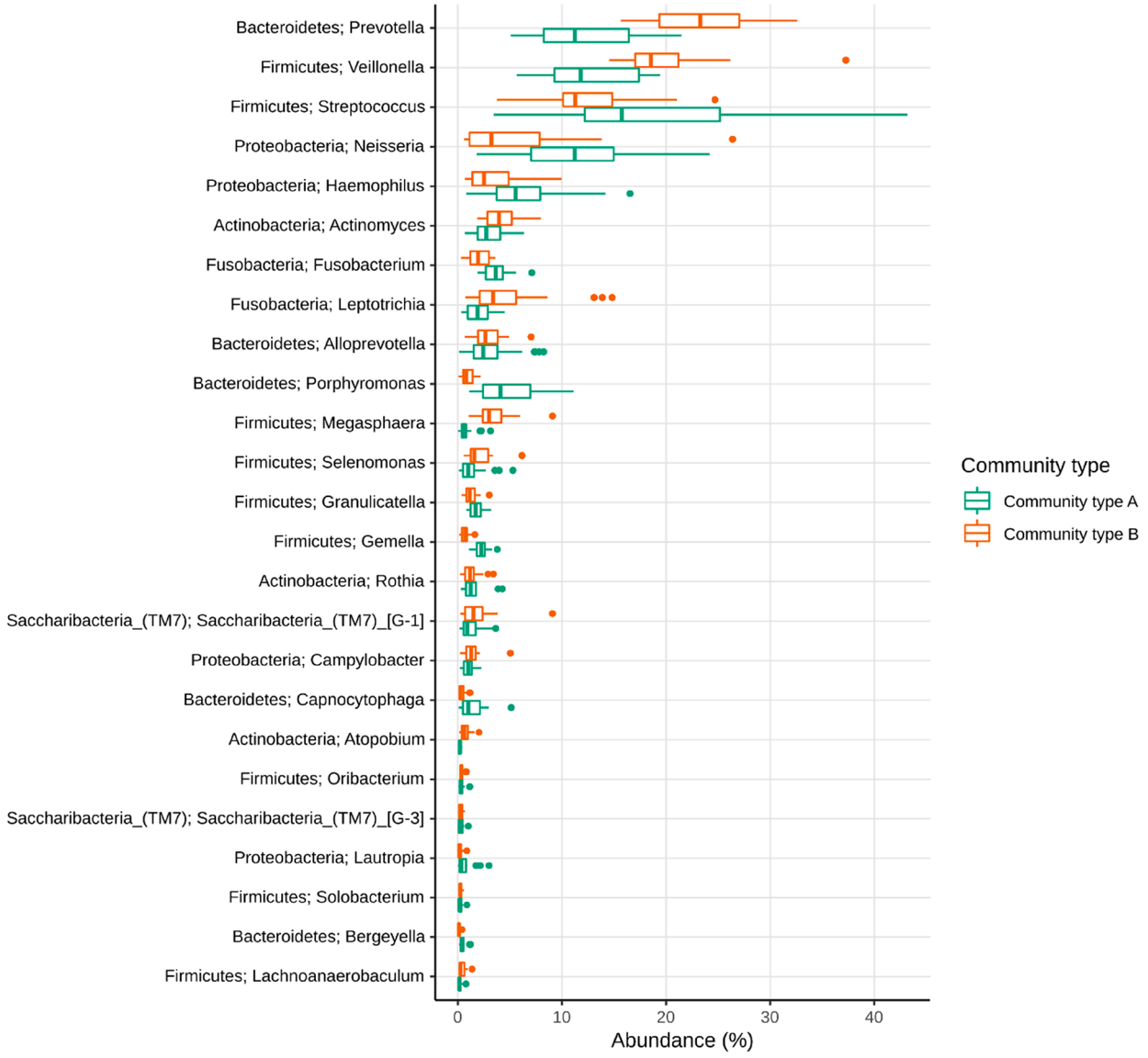


Figure 1. Box plot showing the abundance distribution of the top genera in 41 samples grouped by community type.

guaiacyl glucoside and guaiacol between guaiacyl glucoside tasters (n = 16) and non-tasters (n = 7).

RESULTS AND DISCUSSION

Defining Sensory Phenotype. Four phenotypes were observed among the 41 individuals based on their ability to perceive fruity/floral flavor from aqueous geranyl glucoside and smoky/medicinal flavor from guaiacyl glucoside solutions, with proportions comparable to a similar study conducted in 2015.[6] Of the 41 individuals tested in this study, 49% perceived significant flavor from both geranyl glucoside and guaiacyl glucoside, 15% perceived significant flavor from neither, 34% perceived significant flavor from guaiacyl glucoside and not geranyl glucoside, and only 2% (1 person) perceived significant flavor from geranyl glucoside and not guaiacyl glucoside (Table S1 of the Supporting Information). Overall, 51% had a significant rating of fruity/floral flavor for geranyl glucoside, and 83% had a significant rating of smoky/medicinal flavor for guaiacyl glucoside. The concentration used in the test was selected on the basis of prior experience, with the observation that some individuals perceived no flavor and others responded strongly. On the basis of these data, 23 individuals were selected for further testing, representing the four sensory phenotypes: "taster" of both glucosides, "non-taster" of both glucosides, "taster" of geranyl glucoside but "non-taster" of guaiacyl glucoside, and "non-taster" of geranyl glucoside and "taster" of guaiacyl glucoside.

Stability of Glucoside Response over Time. Over an 8 year period of testing, sensory responses to glucosides were observed to be quite consistent. See Table S1 of the Supporting Information for 2018 individual results. A total of 57 individuals assessed geranyl glucoside (3.00 mg/L) and guaiacyl glucoside (0.500 mg/L) on four occasions between 2010 and 2018, using similar testing protocols, and 35 people participated in the test more than once (data not shown). Around 50% of the individuals responded to geranyl glucoside each time, and 50−83% of the individuals responded to guaiacyl glucoside, increasing with familiarity with the task and improvements to the sensory protocol. Individual results were largely consistent (69%; data not shown) over the period of testing, with the variation probably reflecting the complexity of

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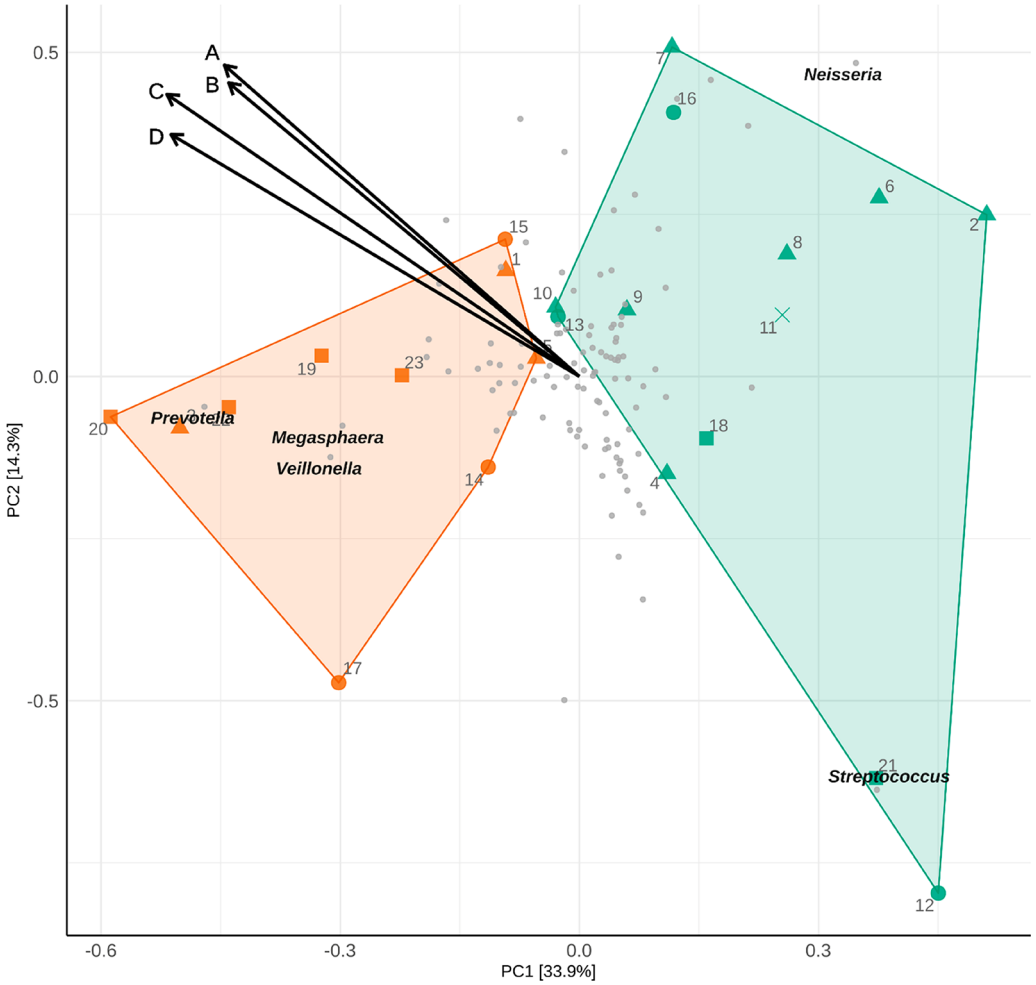


Figure 2. PCA of the Hellinger-transformed OTU abundances of 23 samples. Samples are color-framed by community type (green, A; orange, B) and the 5 most influential OTUs are labeled to the genus level. Symbols denote sensory phenotypes: (▲) GerGlc and GuGlc, (×) GerGlc only, (●) GuGlc only, and (■) no response to either glucoside, and numbers identify the 23 individuals in the study. OTUs are shown as gray dots. Saliva hydrolysis values (significance $p < 0.05$) were fitted into the plot using the envfit function in R. A, guaiacyl glucoside hydrolysis; B, geranyl glucoside hydrolysis; C, guaiacol released; and D, geraniol released.

the biological system of the mouth, in addition to expected day-to-day variation in odorant perception acuity.

Saliva Microbial Community Profiling. The bacterial microbial populations present in the oral cavity were assessed by collecting and sequencing DNA from the saliva of the 41 individuals. After bioinformatic processing, a total of 363 OTUs with bacterial classification were obtained. Within the combined data set, 10 bacterial phyla were observed, from which Firmicutes (43.8%), Bacteroidetes (25.3%), Proteobacteria (16.9%), Fusobacteria (6.6%), and Actinobacteria (5.9%) were the most abundant (Figure S1 of the Supporting Information). These bacterial phyla were mainly represented by OTUs classified to the genus level as *Prevotella* (17.8%), *Streptococcus* (16.8%), *Veillonella* (16.3%), *Neisseria* (8.7%), and *Haemophilus* (4.9%) (Figure 1 and Figure S1 of the Supporting Information). Several metagenomic studies have observed similar composition of dominant phyla as seen in this study, corroborating the concept of a core oral microbiota,[18,21,22,41] albeit with a large degree of interindividual variation of microbial community structure, attributed to host-related biochemical parameters of the saliva and states of dysbiosis.[18]

Heterogeneity between individual oral microbiota existed within the 41 samples. Two community types were identified by applying the Dirichlet multinomial mixture model (DMM) analysis on the OTU abundances. Community type A showed the highest proportions of OTUs belonging to the genera *Streptococcus*, *Neisseria*, *Haemophilus*, and *Poryphyromonas*, whereas community type B was represented by higher abundance of *Prevotella* and *Veillonella* (Figure 1). The largest study performed to date on the salivary microbiome of 2344 individuals reported similar results,[21] with two co-habitating community types: type I characterized by the dominance of mainly *Prevotella*, *Veillonella*, *Actinomyces*, *Rothia*, *Streptococcus salivarius*, and *Streptococcus parasanguinis*, and type II characterized by higher abundance of *Neisseria*, *Haemophilus*, *Porphyromonas*, *Gemella*, and *Streptococcus mitis*. The observed similarities between the microbial community structure suggest that the 41 individuals sampled for this study are representative of a larger community size.

A subset of 23 individuals representing both community types and the four phenotypes of sensory responses to the glucosides were tested for saliva hydrolysis of geranyl glucoside and guaiacyl glucoside, and the relationships with the saliva microbiome were investigated. The salivary microbiome and

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community type classification was consistent between two replicate samples obtained within a 4 week interval for the subset of the 23 individuals selected for further sensory and saliva tests. Hierarchical clustering analyses of the OTU abundance between time points revealed a high conservation of the microbial community structure throughout time, and only three samples (individuals 8, 9, and 13) did not cluster together across time points (Figure S2 of the Supporting Information). These results are consistent with a previous study, where the microbial community of 5 individuals was stable throughout a period of a week.[19]

Relationships between the Saliva Microbiome and *In Vitro* Hydrolysis of Glucosides with Whole Fresh Saliva. Principal component analyses (PCAs) of the salivary microbiome of the subset of 23 individuals confirmed the heterogeneity between samples and clustering based on DMM community types (Figure 2). OTUs classified as *Prevotella*, *Streptococcus*, *Neisseria*, *Veillonella*, and *Megasphaera* were the most influential in the PCAs (Figure 2) as well as differentially abundant between community types A and B (Figure 1).

Community type was associated with *in vitro* hydrolysis of glucosides in saliva. The release of geraniol from geranyl glucoside varied from 5 to 81%, and the release of guaiacol from guaiacyl glucoside ranged from 2 to 94%. These results have a wider range than the 10−40% of hydrolysis during 2−5 min in the mouth reported in the literature.[5,10] The concentrations of remaining geranyl glucoside and guaiacyl glucoside following incubation with saliva were also quantified to allow for a direct calculation of hydrolyzed glucoside, and these were highly correlated with hydrolysis calculated by measuring the release of the free volatiles (Table S2 of the Supporting Information). The percentages of geraniol and guaiacol release by hydrolysis *in vitro* were fitted into the ordination plot using the envfit function (Figure 2). All measurements were significantly correlated ($p < 0.05$) with principal component 1 (PC1) and principal component 2 (PC2), revealing an association between the saliva hydrolysis results and bacterial communities. Interestingly, OTUs abundant in individuals from community type B in the orange frame were largely associated with higher hydrolysis of glucosides. To further investigate this association, correlation analysis was performed between hydrolysis measurements and the abundance of the top 50 OTUs. Significant correlations [$p < 0.05$; false discovery rate (FDR) corrected] were observed for 11 OTUs, from which specific *Streptococcus* and *Prevotella* OTUs showed high correlation coefficients (r values of 0.93 and 0.64, respectively; Figure 3). In general, genera abundant in community types A and B showed negative and positive correlations with the hydrolysis of glucosides, respectively (Figure 3). Furthermore, no specificity to a particular glucoside was observed, with both geranyl and guaiacyl glucosides showing the same correlation patterns. These results indicate that specific bacterial communities are responsible for the interindividual variation of saliva hydrolysis of glucosides and help to explain differences between individuals reported in the saliva hydrolysis of flavonoid and grape-derived glucosides.[8,23] It is unclear whether a specific species is responsible for the production of extracellular β-glucosidases in the saliva, because this common trait is shared throughout different genera of the oral microbiota, including *Streptococcus*, *Prevotella*, *Actinomyces*, *Granulicatella*, and *Fusobacterium*.[16,23,42] Nevertheless, the correlation results suggest a multi-species effect, with each

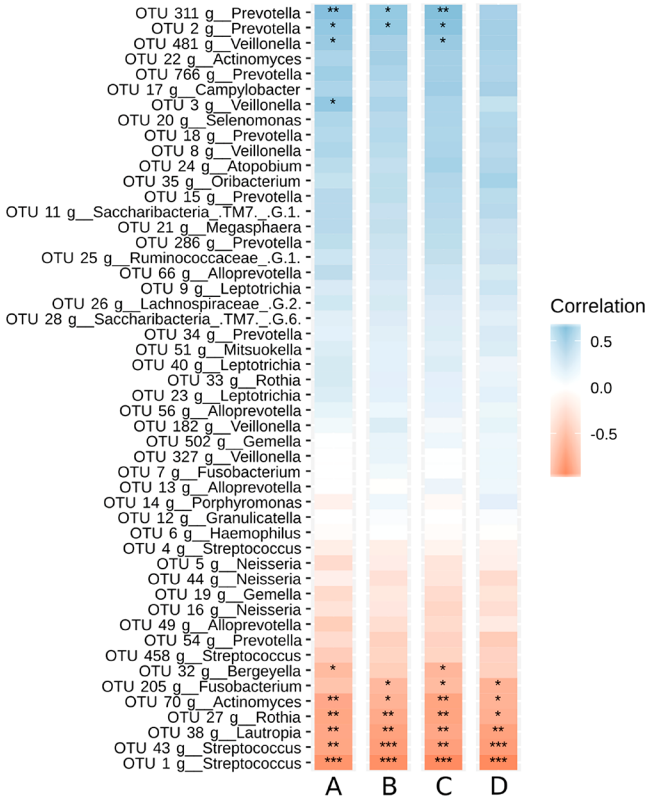


Figure 3. Pearson correlation coefficient results of the top 50 OTUs with the saliva hydrolysis percentage values. Genus classification for individual OTUs is provided. Benjamini−Hochberg-corrected p value significance levels are represented as asterisks (***, <0.001; **, <0.01; and *, <0.05). A, release of geraniol; B, hydrolysis of geranyl glucoside; C, release of guaiacol; and D, hydrolysis of guaiacyl glucoside.

member contributing to the hydrolysis of glucosides rather than one specific species. Deeper taxonomic assignment with longer 16S rRNA gene sequencing would allow for association of the OTUs with saliva bacterial genomes[24] and investigation of the presence of glucosidase genes. Salivary chemical composition parameters (e.g., pH, albumin, chitinase, mucins, and lysozyme levels) must also be taken into consideration, because they have been associated with the different microbial community types[18] and could be responsible for the activity of β-glucosidases as well as influence the volatility of aroma compounds.[43,44]

The relationship between the salivary microbiome, phenotype, and sensory detection thresholds (see below) of the 23 individuals were investigated using the LDA effective size (LEfSe) biomarker discovery tool and correlation analyses. LEfSe revealed that none of the four phenotypes were characterized by an abundance of relevant OTUs. Furthermore, no significant correlations were observed between the sensory thresholds (see later for sensory thresholds) and the abundance of specific OTUs (data not shown). No clustering based on phenotype was observed in PCA analyses of the microbiome; in fact, the salivary microbiome of individuals with phenotypes unable to taste both or one glucoside formed part of community type B (Figure 3), characterized by bacterial communities correlated with higher saliva hydrolysis.

Relationships between Saliva Hydrolysis and Sensory Phenotype. If low hydrolytic activity toward specific

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glucosides is limiting flavor perception, there should be significant differences in hydrolysis between tasters and non-tasters for that particular glucoside. Furthermore, for those phenotypes consisting of flavor perception from only one of the two glucosides, the hydrolysis of the two glucosides should vary, reflecting the phenotype. However, when considering each glucoside separately for the 23 individuals, there was no significant difference (ANOVA at 95% confidence) between the geranyl glucoside tasters and non-tasters in their salivary release of geraniol (p = 0.478) and no significant difference between guaiacyl glucoside tasters and non-tasters in their salivary release of guaiacol (p = 0.483). The results indicate that the saliva glycosidase activity was not limiting glucoside flavor release. The release of geraniol from geranyl glucoside varied from 5 to 81%, and release of guaiacol from guaiacyl glucoside ranged from 2 to 94%. These results are somewhat higher but comparable to the 10−40% of hydrolysis during 2−5 min in the mouth reported in the literature.[5,10] There was a strong correlation between the release of geraniol and guaiacol (r = 0.883; p < 0.001), suggesting that the hydrolysis was not the limiting factor for those phenotypes consisting of flavor perception from only one of the two glucosides. The concentration of remaining geranyl glucoside and guaiacyl glucoside following incubation with saliva was also quantified to allow for a direct calculation of the hydrolyzed glucoside, and these were highly correlated with the hydrolysis calculated by measuring the release of the free volatiles (Table S2 of the Supporting Information).

The key factors that can influence flavor perception from glucosides are expected to be the oral microbiota, the hydrolysis of glucosides in saliva, and the detection of the released volatile odorant. The lack of relationship between the saliva hydrolysis and the sensory phenotype indicates that the salivary glycosidase activity was not a major factor determining the ability to perceive flavor from glucosides. It is possible that the results obtained after 30 min of incubation are influenced by further metabolism of geraniol and guaiacol and do not fully reflect the hydrolysis that can occur *in vivo* in the first 2 min. The saliva hydrolysis kinetics over the first 2 min would be a challenging and interesting aspect to investigate further.

Overall, the lack of relationships between the microbial communities, hydrolysis of glucosides, and sensory data suggest that the oral microbiota is not the major factor driving the flavor perception of glucosides as previously hypothesized; instead, the perception of the released odorant could be more important than previously thought.[23]

Sensory Detection Thresholds of Glucosides and Volatile Odorants. The 23 individuals recruited for threshold testing represented the four sensory phenotypes and the two oral microbiota communities. The distribution of all individual thresholds is summarized in Figure 4.

The two glucosides were perceptible by most of the individuals, in some cases at surprisingly low concentrations. The panel BET for geranyl glucoside was determined to be 5080 nmol/L or 1610 μg/L (Figure 4 and Table S2 of the Supporting Information), which is below the concentration used for defining the sensory phenotype (9480 nmol/L). A range of 4 orders of magnitude was observed, with 10 individuals having an estimated threshold above the maximum concentration tested. Note that the lack of material prevented retesting at a higher concentration. Three individuals had very low thresholds for geranyl glucoside, with two at 17.3 nmol/L and one at 1.92 nmol/L. The high proportion of individuals

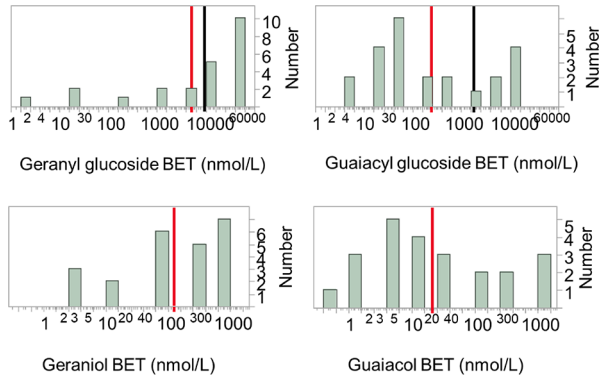


Figure 4. Retronasal detection BET frequency histograms (n = 23) for geranyl glucoside, guaiacyl glucoside, geraniol, and guaiacol in water, projected on the $\log_{10}$ scale with the concentration expressed in nanomoles per liter to assist comparisons between compounds of different molecular masses. Red lines represent the panel BETs, and black lines mark the concentrations of the glucosides used to define sensory phenotype.

with high geranyl glucoside thresholds is probably related to the number of people with high thresholds for geraniol (n = 7; 30%), slightly higher compared to averages of 14−23% observed by Flaherty and Lim.[12] The panel BET for geraniol was found to be 121 nmol/L, which is somewhat higher than orthonasal thresholds reported in the literature.[45−47] This is reasonable given that different neurological pathways are used during orthonasal and retronasal sensing and trigeminal and taste influence the intra-oral retronasal mode.[48,49] The distribution mirrors the distribution of thresholds for geranyl glucoside, albeit with an order of magnitude smaller range.

The panel BET for guaiacyl glucoside was determined to be 240 nmol/L or 68.6 μg/L, well below the concentration used for defining the sensory phenotype (1750 nmol/L). A range of 4 orders of magnitude was observed, with four individuals having a threshold above the concentration tested. Six individuals had very low thresholds for guaiacyl glucoside, with four at 20.2 nmol/L and two at 6.05 nmol/L. The threshold for guaiacol was determined to be 21.2 nmol/L, which is slightly higher but comparable to thresholds reported in the literature.[45,50] The distribution mirrors the distribution of thresholds for guaiacyl glucoside, albeit with an order of magnitude smaller range.

Some individuals were able to detect glucoside at an equimolar concentration to that of aglycone (Table S2 of the Supporting Information). While the threshold testing was all performed by in-mouth retronasal assessment, the panel was not trained in a prescribed protocol of mouth movements and breathing; therefore, this might still play a role in the ability of some to taste the glucosides. In fact, one of the most sensitive individuals reported doing a self-taught special technique for retronasal tasting similar to that described by Buettner.[51] If the inability to taste the glucoside was due to the absence of salivary enzymes, we would expect to see a distinct bimodal distribution of the thresholds for the glucosides, which was not observed.

The thresholds for individuals are presented in Table S2 of the Supporting Information, along with a comment on the reliability of the data of the individual based on the consistency of the data between the screening test and the threshold test, sensory experience of the individual, amount of noise in the data, and individual comments from the threshold test to

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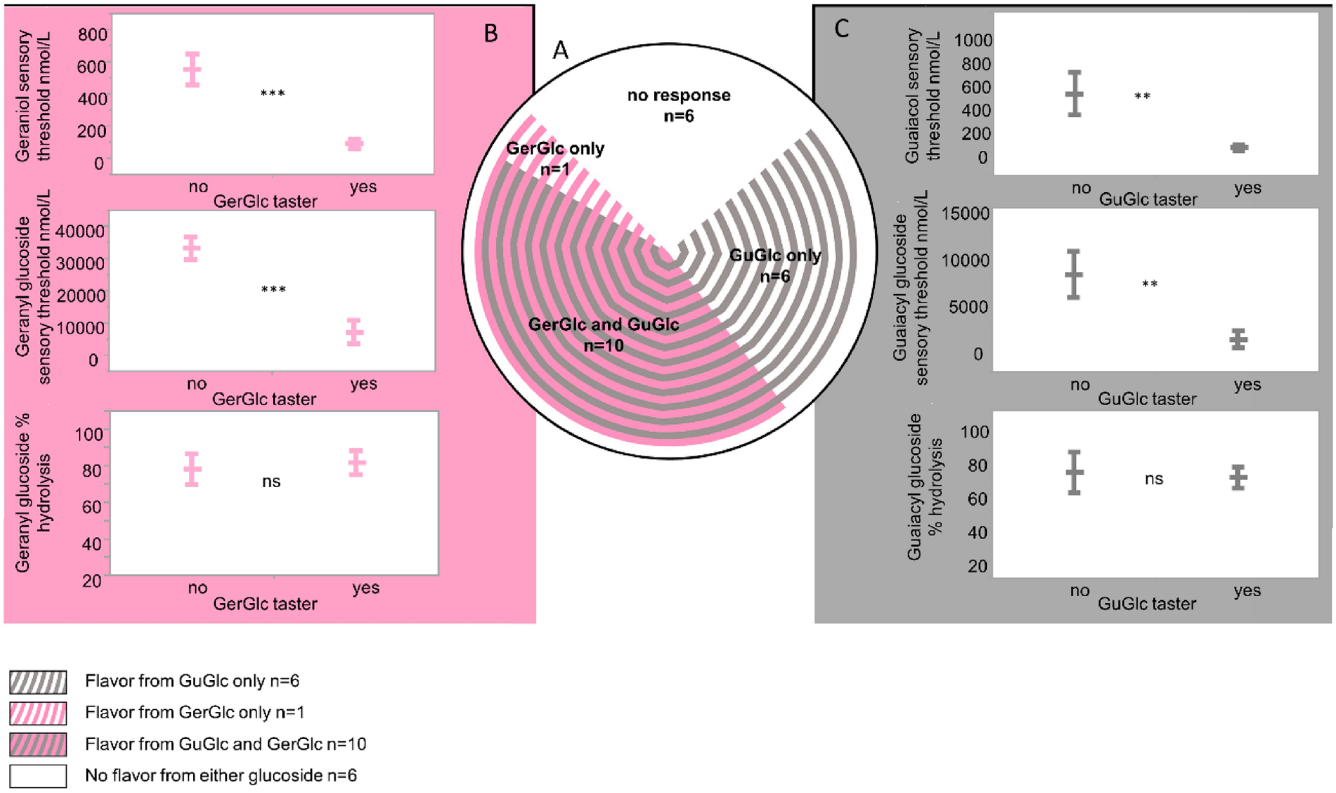


Figure 5. (A) Sensory phenotype overview of 23 individuals based on flavor perceived from geranyl glucoside (GerGlc, 9480 nmol/L) and guaiacyl glucoside (GuGlc, 1750 nmol/L) in water. Individuals who perceived fruity/floral flavor from geranyl glucoside are represented by pink lines (n = 11), and those who perceived smoky/medicinal flavor from guaiacyl glucoside are represented by gray lines (n = 16). Individuals who perceived flavor from both glucosides (GerGlc and GuGlc) are represented by pink and gray lines (n = 10), and individuals who did not perceive flavor from either glucoside (NR = no response) are represented by the white area (n = 6). (B) Comparison of mean values for geraniol sensory detection threshold, geranyl glucoside sensory detection threshold, and geranyl glucoside percentage of hydrolysis during *in vitro* incubation with whole fresh saliva of groups of geranyl glucoside (GerGlc) tasters (n = 11) and non-tasters (n = 12) as defined in panel A. Error bars indicate standard error. (C) Comparison of mean values for guaiacol sensory detection threshold, guaiacyl glucoside sensory detection threshold, and guaiacyl glucoside percentage of hydrolysis during *in vitro* incubation with whole fresh saliva of groups of guaiacyl glucoside (GuGlc) tasters (n = 16) and non-tasters (n = 7) as defined in panel A. Error bars indicate standard error. (***) $p < 0.001$, and (**) $p < 0.01$. ns = not significant.

indicate guesses or confident flavor recognition. Individual results should not be overinterpreted,[52] and fatigue may have introduced extra error into these data, because panellists were required to spend up to 3 h in testing on 1 day, although with extensive breaks. Consequently, some individual thresholds were likely erroneously low as a result of guessing, and others may have been erroneously high as a result of fatigue and lack of attention toward the end of the sensory session. Rather than carefully considering individual responses, a more rigorous approach is to compare the thresholds of groups containing seven or more individuals.

Relationships between Thresholds and Sensory Phenotype. Glucoside thresholds were significantly different between the glucoside tasters and the non-tasters. Tasters of geranyl glucoside (n = 11) had a significantly lower ($p < 0.001$) geranyl glucoside threshold than non-tasters (n = 12), and tasters of guaiacyl glucoside (n = 16) had a significantly lower ($p < 0.05$) guaiacyl glucoside threshold than non-tasters (n = 7). There was a spread of thresholds within the groups of tasters and non-tasters rather than two completely discrete groups. Accordingly, it might be more appropriate to describe the distribution of thresholds along a continuum, similar to the distribution of orthonasal thresholds, rather than in binary categories, such as "taster" and "non-taster" phenotypes (Figure 5).

Overall, odorant detection threshold was a good predictor of the ability to perceive flavor from glucosides. The thresholds of the free volatiles, geraniol and guaiacol, were significantly different between the phenotypes. Tasters of geranyl glucoside (n = 11) had significantly lower geraniol thresholds than non-tasters (n = 12) ($p < 0.001$), and tasters of guaiacyl glucoside (n = 16) had significantly lower guaiacol thresholds than non-tasters (n = 7) ($p < 0.05$).

In contrast, there was no significant difference between tasters and non-tasters in the salivary hydrolysis of either of the two glucosides or the microbial communities as described above.

To our knowledge this study has been the first to investigate oral microbiota, saliva hydrolysis, and retronasal sensory detection thresholds with a large number of individuals, to understand interindividual variation in perception of flavor from glucoside hydrolysis in-mouth. The oral microbiota varied between individuals, and two main clusters of microbial communities were observed, characterized by differences in relative abundances of OTUs of the genera *Streptococcus*, *Neisseria*, *Haemophilus*, *Poryphyromonas*, *Prevotella*, and *Veillonella*. The *Prevotella* and *Veillonella* dominant community had higher glycosidase activity in saliva; however, this link did not relate to sensory phenotypes. While oral microbiota are vitally important to flavor release from glucosides in-mouth, being the

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major source of glycosidase enzymes, this was not limiting flavor perception from glucosides in our study. Counter to our expectations, the retronasal detection threshold of the volatile odorant and not saliva hydrolysis was most strongly related to sensory phenotype. The two glucosides were perceptible by most of the individuals, in some cases at surprisingly low concentrations. The data suggest that the ability to detect flavor from glucosides is not a simple binary yes/no system limited by enzyme glycosidase activity but rather that there is a distribution of sensory thresholds for the glucosides, which follows the distribution of the thresholds for the free volatiles. It then follows that most people should be able to detect flavor from glucosides if they can perceive aglycone and the concentration of glucoside is above the threshold for that person. This means that glucosides might be capable of contributing to flavor release in-mouth for most people so long as the concentration is high enough. In comparison of these results to the context of wine, monoterpene glycosides in aromatic varieties, such as Gewürztraminer wines, are often in the range of 0.5−1.0 mg/L.[6] Surprisingly, concentrations below 20 nmol/L could be perceived by a small proportion of highly sensitive individuals, indicating that very low concentrations of glucosides can contribute to flavor through in-mouth release.

More work is needed to characterize the links between microbiota, quantitative and qualitative differences in salivary enzymes, physicochemical properties of saliva, and rapid kinetics during the time frame of tasting with a broader range of substrates. It would be interesting to test more glycosides, from simple glucosides of various volatile odorants and tastants to more complex disaccharides commonly found in foods, both individually and in mixtures, and test the flavor and taste impacts of glycoside hydrolysis in complex food matrices.

The findings about the plasticity of sensory responses to glycosides can also guide translation of results from sensory model experiments under controlled conditions to real world consumer sensory experiences. If the factor limiting flavor perception from glycosides was hydrolysis determined by oral microbiota, there might be opportunities to improve the ability to taste glycosides by deliberately changing the oral microbiota, for example by increasing live cultures and prebiotics in the diet. Our data would suggest that this would not dramatically improve flavor release from glycosides, although further work would be needed to rule this out. Conversely, use of antibiotic mouthwash would be expected to decrease sensitivity to glycosides by inhibiting hydrolysis. For sensory professionals, training in the retronasal tasting technique might increase sensitivity to glycosides as one sensitive individual reported using a self-taught retronasal tasting technique.

Overall, this study has shown that, even though oral microbiota determine saliva hydrolysis, saliva hydrolysis of geranyl and guaiacyl glucosides is not the main factor limiting perception of retronasal odor from glucoside hydrolysis in-mouth; instead, the odorant threshold is the main factor determining flavor perception from these glucosides.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jafc.9b05450.

Method details for *in vitro* saliva hydrolysis, heatmaps representing the percentage read abundance of OTUs grouped by (A) phylum and (B) genus from saliva samples of 41 individuals (Figure S1), temporal variation of oral microbiota between two saliva samples from the same individual (n = 23) (Figure S2), phenotype category and mean scores for each judge for "fruity/floral" and "smoky/medicinal" flavor attributes (n = 3), together with the significance level from one-way ANOVAs, for geraniol glucoside (GerGlc, 9480 nmol/L), guaiacol glucoside (GuGlc, 1750 nmol/L), and the water control (control), with significant responses shown in bold and judges numbered 1−23 selected for threshold testing (Table S1), sensory results and *in vitro* saliva hydrolysis results for 23 individuals, including tasting phenotype defined by assessing geranyl glucoside (3.00 mg/L) and guaiacyl glucoside (0.500 mg/L) in water, saliva hydrolysis calculated by the concentration of free volatile and glucoside, and retronasal detection thresholds of geraniol and geranyl glucosides, guaiacol, and guaiacyl glucoside (Table S2) (PDF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Cordonnier, R.; Bayonove, C. Bound Monoterpenes in the Grape Berry of Muscat of Alexandria Released by One or More Enzymes from the Fruit. (Biochimie Appliquee. Mise En Evidence Dans La Baie de Raisin, Variete Muscat d'Alexandrie, de Monoterpenes Lies Revelables Par Une Ou Plusieurs). *C. R. Acad. Sci. Paris* **1974**, 3387−3390.

(2) Williams, P. J.; Strauss, C. R.; Wilson, B.; Massy-Westropp, R. A. Novel Monoterpene Disaccharide Glycosides of *Vitis vinifera* Grapes and Wines. *Phytochemistry* **1982**, *21* (8), 2013−2020.

DOI: 10.1021/acs.jafc.9b05450
J. Agric. Food Chem. 2020, 68, 10299−10309

(3) Williams, P. J.; Strauss, C. R.; Wilson, B. Classification of the Monoterpenoid Composition of Muscat Grapes. *Am. J. Enol. Vitic.* **1981**, *32* (3), 230−235.
(4) Starkenmann, C.; Le Calve, B.; Niclass, Y.; Cayeux, I.; Beccucci, S.; Troccaz, M. Olfactory perception of cysteine-*S*-conjugates from fruits and vegetables. *J. Agric. Food Chem.* **2008**, *56* (20), 9575−9580.
(5) Hemingway, K. M.; Alston, M. J.; Chappell, C. G.; Taylor, A. J. Carbohydrate-flavour conjugates in wine. *Carbohydr. Polym.* **1999**, *38* (3), 283−286.
(6) Parker, M.; Black, C. A.; Barker, A.; Pearson, W.; Hayasaka, Y.; Francis, I. L. The contribution of wine-derived monoterpene glycosides to retronasal odour during tasting. *Food Chem.* **2017**, *232*, 413−424.
(7) Parker, M.; Osidacz, P.; Baldock, G. A.; Hayasaka, Y.; Black, C. A.; Pardon, K. H.; Jeffery, D. W.; Geue, J. P.; Herderich, M. J.; Francis, I. L. Contribution of several volatile phenols and their glycoconjugates to smoke-related sensory properties of red wine. *J. Agric. Food Chem.* **2012**, *60* (10), 2629−2637.
(8) Walle, T.; Browning, A. M.; Steed, L. L.; Reed, S. G.; Walle, U. K. Flavonoid glucosides are hydrolyzed and thus activated in the oral cavity in humans. *J. Nutr.* **2005**, *135* (1), 48−52.
(9) Mallery, S. R.; Budendorf, D. E.; Larsen, M. P.; Pei, P.; Tong, M.; Holpuch, A. S.; Larsen, P. E.; Stoner, G. D.; Fields, H. W.; Chan, K. K.; Ling, Y.; Liu, Z. Effects of human oral mucosal tissue, saliva, and oral microflora on intraoral metabolism and bioactivation of black raspberry anthocyanins. *Cancer Prev. Res.* **2011**, *4* (8), 1209−1221.
(10) Kamonpatana, K.; Failla, M. L.; Kumar, P. S.; Giusti, M. M. Anthocyanin structure determines susceptibility to microbial degradation and bioavailability to the buccal mucosa. *J. Agric. Food Chem.* **2014**, *62* (29), 6903−10.
(11) Stradwick, L.; Inglis, D.; Kelly, J.; Pickering, G. Development and application of assay for determining β-glucosidase activity in human saliva. *Flavour* **2017**, *6* (1), 1−8.
(12) Flaherty, T. J.; Lim, J. Individual Differences in Retronasal Odor Responsiveness: Effects of Aging and Concurrent Taste. *Chemosens. Percept.* **2017**, *10* (4), 91−103.
(13) Buettner, A.; Beauchamp, J. Chemical input − Sensory output: Diverse modes of physiology−flavour interaction. *Food Qual. Prefer.* **2010**, *21* (8), 915−924.
(14) Chauncey, H. H.; Lionetti, F.; Winer, R. A.; Lisanti, V. F. Enzymes of human saliva. I. The determination, distribution, and origin of whole saliva enzymes. *J. Dent. Res.* **1954**, *33* (3), 321−34.
(15) van der Hoeven, J. S.; van den Kieboom, C. W.; Camp, P. J. Utilization of mucin by oral Streptococcus species. *Antonie van Leeuwenhoek* **1990**, *57* (3), 165−72.
(16) Sakamoto, M.; Suzuki, M.; Huang, Y.; Umeda, M.; Ishikawa, I.; Benno, Y. *Prevotella shahii* sp. nov. and *Prevotella salivae* sp. nov., isolated from the human oral cavity. *Int. J. Syst. Evol. Microbiol.* **2004**, *54* (3), 877−883.
(17) Dewhirst, F. E.; Chen, T.; Izard, J.; Paster, B. J.; Tanner, A. C.; Yu, W. H.; Lakshmanan, A.; Wade, W. G. The human oral microbiome. *J. Bacteriol.* **2010**, *192* (19), 5002−17.
(18) Zaura, E.; Brandt, B. W.; Prodan, A.; Teixeira de Mattos, M. J.; Imangaliyev, S.; Kool, J.; Buijs, M. J.; Jagers, F. L.; Hennequin-Hoenderdos, N. L.; Slot, D. E.; Nicu, E. A.; Lagerweij, M. D.; Janus, M. M.; Fernandez-Gutierrez, M. M.; Levin, E.; Krom, B. P.; Brand, H. S.; Veerman, E. C.; Kleerebezem, M.; Loos, B. G.; van der Weijden, G. A.; Crielaard, W.; Keijser, B. J. On the ecosystemic network of saliva in healthy young adults. *ISME J.* **2017**, *11* (5), 1218−1231.
(19) Belstrom, D.; Holmstrup, P.; Bardow, A.; Kokaras, A.; Fiehn, N. E.; Paster, B. J. Temporal Stability of the Salivary Microbiota in Oral Health. *PLoS One* **2016**, *11* (1), No. e0147472.
(20) Zhou, Y.; Gao, H.; Mihindukulasuriya, K. A.; La Rosa, P. S.; Wylie, K. M.; Vishnivetskaya, T.; Podar, M.; Warner, B.; Tarr, P. I.; Nelson, D. E.; Fortenberry, J. D.; Holland, M. J.; Burr, S. E.; Shannon, W. D.; Sodergren, E.; Weinstock, G. M. Biogeography of the ecosystems of the healthy human body. *Genome Biol.* **2013**, *14* (1), R1.
(21) Takeshita, T.; Kageyama, S.; Furuta, M.; Tsuboi, H.; Takeuchi, K.; Shibata, Y.; Shimazaki, Y.; Akifusa, S.; Ninomiya, T.; Kiyohara, Y.; Yamashita, Y. Bacterial diversity in saliva and oral health-related conditions: The Hisayama Study. *Sci. Rep.* **2016**, *6* (1), 22164.
(22) De Filippis, F.; Vannini, L.; La Storia, A.; Laghi, L.; Piombino, P.; Stellato, G.; Serrazanetti, D. I.; Gozzi, G.; Turroni, S.; Ferrocino, I.; Lazzi, C.; Di Cagno, R.; Gobbetti, M.; Ercolini, D. The same microbiota and a potentially discriminant metabolome in the saliva of omnivore, ovo-lacto-vegetarian and Vegan individuals. *PLoS One* **2014**, *9* (11), No. e112373.
(23) Muñoz-González, C.; Cueva, C.; Ángeles Pozo-Bayón, M.; Victoria Moreno-Arribas, M. Ability of human oral microbiota to produce wine odorant aglycones from odourless grape glycosidic aroma precursors. *Food Chem.* **2015**, *187*, 112−119.
(24) Escapa, I. F.; Chen, T.; Huang, Y.; Gajare, P.; Dewhirst, F. E.; Lemon, K. P. New Insights into Human Nostril Microbiome from the Expanded Human Oral Microbiome Database (eHOMD): A Resource for the Microbiome of the Human Aerodigestive Tract. *mSystems* **2018**, *3* (6), e00187-18.
(25) Avila, M.; Ojcius, D. M.; Yilmaz, O. The oral microbiota: Living with a permanent guest. *DNA Cell Biol.* **2009**, *28* (8), 405−411.
(26) Mayr, C. M.; Parker, M.; Baldock, G. A.; Black, C. A.; Pardon, K. H.; Williamson, P. O.; Herderich, M. J.; Francis, I. L. Determination of the Importance of In-Mouth Release of Volatile Phenol Glycoconjugates to the Flavor of Smoke-Tainted Wines. *J. Agric. Food Chem.* **2014**, *62* (11), 2327−2336.
(27) Hayasaka, Y.; Dungey, K. A.; Baldock, G. A.; Kennison, K. R.; Wilkinson, K. L. Identification of a beta-D-glucopyranoside precursor to guaiacol in grape juice following grapevine exposure to smoke. *Anal. Chim. Acta* **2010**, *660* (1−2), 143−8.
(28) Pedersen, D. S.; Capone, D. L.; Skouroumounis, G. K.; Pollnitz, A. P.; Sefton, M. A. Quantitative analysis of geraniol, nerol, linalool, and α-terpineol in wine. *Anal. Bioanal. Chem.* **2003**, *375* (4), 517−522.
(29) Pollnitz, A. P.; Pardon, K. H.; Sykes, M.; Sefton, M. A. The effects of sample preparation and gas chromatograph injection techniques on the accuracy of measuring guaiacol, 4-methylguaiacol and other volatile oak compounds in oak extracts by stable isotope dilution analyses. *J. Agric. Food Chem.* **2004**, *52* (11), 3244−52.
(30) Bolger, A. M.; Lohse, M.; Usadel, B. Trimmomatic: A flexible trimmer for Illumina sequence data. *Bioinformatics* **2014**, *30* (15), 2114−20.
(31) Magoc, T.; Salzberg, S. L. FLASH: Fast length adjustment of short reads to improve genome assemblies. *Bioinformatics* **2011**, *27* (21), 2957−63.
(32) Edgar, R. C. UPARSE: Highly accurate OTU sequences from microbial amplicon reads. *Nat. Methods* **2013**, *10* (10), 996−8.
(33) Caporaso, J. G.; Kuczynski, J.; Stombaugh, J.; Bittinger, K.; Bushman, F. D.; Costello, E. K.; Fierer, N.; Peña, A. G.; Goodrich, J. K.; Gordon, J. I.; Huttley, G. A.; Kelley, S. T.; Knights, D.; Koenig, J. E.; Ley, R. E.; Lozupone, C. A.; McDonald, D.; Muegge, B. D.; Pirrung, M.; Reeder, J.; Sevinsky, J. R.; Turnbaugh, P. J.; Walters, W. A.; Widmann, J.; Yatsunenko, T.; Zaneveld, J.; Knight, R. QIIME allows analysis of high-throughput community sequencing data. *Nat. Methods* **2010**, *7*, 335.
(34) Holmes, I.; Harris, K.; Quince, C. Dirichlet multinomial mixtures: Generative models for microbial metagenomics. *PLoS One* **2012**, *7* (2), No. e30126.
(35) R Core Team. *R: A Language and Environment for Statistical Computing*; R Foundation for Statistical Computing: Vienna, Austria, 2015.
(36) Andersen, K. S.; Kirkegaard, R. H.; Karst, S. M.; Albertsen, M. ampvis2: An R package to analyse and visualise 16S rRNA amplicon data. *bioRxiv* **2018**, DOI: 10.1101/299537.
(37) Dixon, P. VEGAN, a package of R functions for community ecology. *J. Veg. Sci.* **2003**, *14* (6), 927−930.
(38) McMurdie, P. J.; Holmes, S. phyloseq: An R Package for Reproducible Interactive Analysis and Graphics of Microbiome Census Data. *PLoS One* **2013**, *8* (4), No. e61217.

DOI: 10.1021/acs.jafc.9b05450
J. Agric. Food Chem. 2020, 68, 10299−10309

(39) Wickham, H. *ggplot2: Elegant Graphics for Data Analysis*; Springer: Cham, Switzerland, 2016; DOI: 10.1007/978-3-319-24277-4.
(40) Segata, N.; Izard, J.; Waldron, L.; Gevers, D.; Miropolsky, L.; Garrett, W. S.; Huttenhower, C. Metagenomic biomarker discovery and explanation. *Genome Biol.* **2011**, *12* (6), R60.
(41) Bik, E. M.; Long, C. D.; Armitage, G. C.; Loomer, P.; Emerson, J.; Mongodin, E. F.; Nelson, K. E.; Gill, S. R.; Fraser-Liggett, C. M.; Relman, D. A. Bacterial diversity in the oral cavity of 10 healthy individuals. *ISME J.* **2010**, *4* (8), 962−74.
(42) Henssge, U.; Do, T.; Radford, D. R.; Gilbert, S. C.; Clark, D.; Beighton, D. Emended description of *Actinomyces naeslundii* and descriptions of *Actinomyces oris* sp. nov. and *Actinomyces johnsonii* sp. nov., previously identified as *Actinomyces naeslundii* genospecies 1, 2 and WVA 963. *Int. J. Syst. Evol. Microbiol.* **2009**, *59* (3), 509−516.
(43) Piombino, P.; Genovese, A.; Esposito, S.; Moio, L.; Cutolo, P. P.; Chambery, A.; Severino, V.; Moneta, E.; Smith, D. P.; Owens, S. M.; Gilbert, J. A.; Ercolini, D. Saliva from obese individuals suppresses the release of aroma compounds from wine. *PLoS One* **2014**, *9* (1), e85611.
(44) Ployon, S.; Morzel, M.; Canon, F. The role of saliva in aroma release and perception. *Food Chem.* **2017**, *226*, 212−220.
(45) Czerny, M.; Christlbauer, M.; Christlbauer, M.; Fischer, A.; Granvogl, M.; Hammer, M.; Hartl, C.; Hernandez, N. M.; Schieberle, P. Re-investigation on odour thresholds of key food aroma compounds and development of an aroma language based on odour qualities of defined aqueous odorant solutions. *Eur. Food Res. Technol.* **2008**, *228* (2), 265−273.
(46) Schuh, C.; Schieberle, P. Characterization of the key aroma compounds in the beverage prepared from Darjeeling black tea: Quantitative differences between tea leaves and infusion. *J. Agric. Food Chem.* **2006**, *54* (3), 916−924.
(47) Ong, P. K. C.; Acree, T. E. Gas chromatography olfactory analysis of lychee (*Litchi chinesis* Sonn.). *J. Agric. Food Chem.* **1998**, *46* (6), 2282−2286.
(48) Bojanowski, V.; Hummel, T. Retronasal perception of odors. *Physiol. Behav.* **2012**, *107* (4), 484−7.
(49) Espinosa Diaz, M. Comparison between orthonasal and retronasal flavour perception at different concentrations. *Flavour Fragrance J.* **2004**, *19* (6), 499−504.
(50) Eisele, T. A.; Semon, M. J. Best estimated aroma and taste detection threshold for guaiacol in water and apple juice. *J. Food Sci.* **2005**, *70* (4), S267−S269.
(51) Buettner, A. Physiology and Chemistry Behind Retronasal Aroma Perception during Winetasting. *Proceedings of the VIIth International Symposium d'Oenology*; Lonvaud-Funel, A.; de Revel, G.; Darriet, P., Eds.; Editions Tec & Doc: Paris, France, 2003; pp 566−569.
(52) Peng, M.; Jaeger, S. R.; Hautus, M. J. Fitting psychometric functions using a fixed-slope parameter: An advanced alternative for estimating odor thresholds with data generated by ASTM E679. *Chem. Senses* **2014**, *39* (3), 229−241.

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